Nan Li

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Education

PhD in Data Science Feb. 2025 - Present

Traffic Medicine, Institute of Forensic Medicine, University of Zurich (UZH) Zurich, Switzerland

- **Research Keywords**: Traffic Medicine, Time-Series Machine Learning, Multi-Modal, Human-Computer Interaction, Explainable Artificial Intelligence, Eye-tracking, EEG, Information Visualization

MSc in Computer Science (Major: Artificial Intelligence) Sep. 2022 - Oct. 2024

University of Zurich (UZH) Zurich, Switzerland

- **Research Keywords**: Explainable Artificial Intelligence, Human-Computer Interaction, Time-Series Classification
- **Master Thesis**: Interpretable Machine Learning Algorithm for Drunk Driving Detection
- **Master Project**: Explainability method for Recommender Systems

BEng in Software Engineering Sep. 2016 - Jun. 2021

Beihang University (BUAA) Beijing, China

- **GPA**: 87/100 (Top 17%)
- **Thesis**: Project Review Assistant System Based on BlockChain

Representative Projects

Multimodal Data Integration and Analysis for Driving Safety Oct. 2024 – Present

Traffic Medicine – Institute of Forensic Medicine – UZH Zurich, Switzerland

- **Objective**: Examine how driving behavior changes when adults with ADHD temporarily stop stimulant medication. Compare medicated and drug-holiday conditions through statistical analysis and interpretable models for driving-state classification and anomaly detection.
- **Contributions**:
  - Lead the implementation of the randomized crossover trial, including participant coordination, study visits, and contributed to experimental design and ethics-related documentation.
  - Used statistical methods including LMM, ANOVA, t-tests to compare repeated-measures clinical and driving outcomes between medicated and drug-holiday conditions.
  - Designed and developed GUI-based tools to support clinical experiments, including experiment setup, device control, automated data recording, synchronization, and anomaly alerts.
  - Developed and maintain a multimodal data synchronization and quality-control pipeline spanning driving-simulator telemetry, eye-tracking, driver video, and EEG data collected across heterogeneous acquisition systems and device clocks.
  - Develop multimodal time-series models for driving-safety classification and anomaly detection.

Multi-Robot Interactive Simulation and Analysis Platform Mar. 2024 - Nov.2024

Programming, Education, and Computer-Human Interaction Lab (PEACH Lab) - ETHz Zurich, Switzerland

- **Objective**: An interactive simulation and visualization framework to facilitate quadrupedal locomotion learning.
- **Contributions**:
  - Formative Study: Conducted interviews with robotics experts to identify key challenges and strategies in locomotion policy analysis.
  - Framework Implementation: Designed and developed the framework, an interactive tool integrating simulation, trajectory, metrics visualization, and key frame capturing for locomotion policy analysis.
  - User Evaluation: Conducted a user study with robotics practitioners to assess the tool's effectiveness, demonstrating its ability to provide immediate insights and enhance exploratory analysis.

Interpretable Machine Learning Algorithm for Drunk Driving Detection Dec. 2023 - Aug.2024

Bosch IoT Lab | ETH Zurich | University of St. Gallen Zurich, Switzerland

- **Objective**: Develop interpretable machine learning algorithms to detect drunk driving using CAN bus data.
- **Contributions**:
  - Analyzed CAN bus and other related data from real vehicles to create accurate models.
  - Developed and trained interpretable time-series classification models including logistic regression, CNNs, RNNs, and state-of-art multivariate time-series classification models.
  - Explored techniques for comprehensive explanations of model predictions.

Explainability Method for Recommender Systems Feb. 2023 - May. 2024

Dynamic and Distributed Information Systems Group - UZH Zurich, Switzerland

- **Objective**: Research and implement explainable methods for recommender system.
- **Contributions**:
  - Extended the Cornac Python package with new explainable recommendation models, explanation algorithms, and evaluation metrics.
  - Developed a pipeline to facilitate the workflow and documented a detailed report.
  - Specialized in matrix factorization-based models and explanation methods.

Publications & Preprints

[1] **Nan Li**, Fan Shi, Xiaoyu Zhang, and April Yi Wang. *MRISA: A Visual Analytics Approach for Comparing Locomotion Policies in Robotics Training.* In Proceedings of Graphics Interface 2025. https://doi.org/10.1145/3769872.3769898

[2] **Nan Li**, Veronika Gambin, Boris Quednow, Pascal Burger, Ana Buadze, Benjamin Offenhammer, Kristina Keller, and Stefan Lakämper. *Impact of drug holidays on simulated driving performance in adults with ADHD: a study protocol for a randomized crossover trial.* Under review at JMIR Research Protocols. http://dx.doi.org/10.21203/rs.3.rs-8611309/v1

[3] Veronika Gambin, **Nan Li**, Esther Irene Schwarz, Kristina Keller, and Stefan Lakämper. *Validating a novel driving simulation-based MWT against the standard MWT in an OSA-cohort challenged by CPAP-withdrawal (DS-MWT2) – Protocol for a monocentric, controlled, randomized, crossover trial. PLOS One*, 21(7):e0341997, 2026. https://doi.org/10.1371/journal.pone.0341997

[4] Tzvetan Popov, **Nan Li**, Veronika Gambin, Kristina Keller, Samuel Wehrli, and Stefan Lakämper. *Predicting driver distraction using a single channel ear EEG.* Available as preprint. https://doi.org/10.64898/2026.01.24.701469

[5] Tzvetan Popov, **Nan Li**, Veronika Gambin, Kristina Keller, Samuel Wehrli, and Stefan Lakämper. *The neuro-ocular costs of texting during driving.* Available as preprint. http://dx.doi.org/10.1101/2025.08.15.670515

Conference Presentations

[1] **Poster**, Sleep Europe: 28th Congress of the European Sleep Research Society (ESRS), Maastricht, Netherlands, Oct. 2026.
[2] **Oral**, 27th World Congress of the International Traffic Medicine Association (ITMA), Tokyo, Japan, Sep. 2026.
[3] **Oral**, Swiss Society of Legal Medicine (SGRM) Annual Meeting, Chur, Switzerland, Jun. 2026.
[4] **Oral**, Swiss Society of Legal Medicine (SGRM) Annual Meeting, Locarno, Switzerland, Jun. 2025.
[5] **Oral**, Graphics Interface 2025, British Columbia, Canada, May 2025.

Industry Experience

Multi-window video player for UAV Aerospace CH UAV Co.,Ltd

QT Application Development May. 2022 - Sep. 2022

- **Main work:** A multi-window video player designed to play both live UDP video streams and local video files, while also incorporating a range of customizable features.
- **Windows application:** Developed using the C++ programming language within the QT framework, this software incorporates a video player powered by ffmpeg.
- **Database:** Utilizing MongoDB to store real-time video data and user-customized information.

Awards and Honors

GRC Travel Grant from UZH for ITMA 2026	Grant	2026
Outstanding Graduate of Beihang University	Honor	Jun. 2021
Excellent Scholarship for Academic Performance (3 times)	Honor	2020, 2019, 2018
Excellent Scholarship for Social Work (2 times)	Scholarship	2019, 2018
Merit Student (2 times)	Honor	2019, 2017
Lee Kum Kee Innovation Scholarship	Scholarship	Jan. 2019
National Undergraduate Innovation and Entrepreneurship Training Program	Grant (Project Lead)	2017-2018

Teaching Experience

Programming in Biology	Tutor	Fall 2025
Foundations of Data Science (Graduate)	Tutor	Fall 2023
Compiler Theory (Undergraduate)	Teaching Assistant	Fall 2020
Object-Oriented Programming (Java) (Undergraduate)	Lead Teaching Assistant	Spring 2020
Algorithm Analysis and Design (Undergraduate)	Teaching Assistant	Fall 2019

Technical Skills

Programming Languages	Python, Java, C++, C, JavaScript, TypeScript, SQL, Kotlin, Golang, Matlab
Data Engineering	Multimodal data synchronization, automated data acquisition, quality control, anonymization
Machine Learning	Time-series classification, explainable artificial intelligence, PyTorch, TensorFlow
Software Development and tools	Git, React, Vue, QT, Spring Boot, MySQL, MongoDB, SolidWorks, LaTeX
Languages	Chinese (Native), English (C1)